Quantum Gravity Dynamics

Chapter 3: Newtonian Calibration of α_G,
the Dirac Spinor, and the Gravitational Potential

A proof-audited working manuscript

Research manuscript — Chapters 1 and 2 preserved

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Abstract

Quantum Gravity Dynamics (QGD) describes effective gravitation through natural-invariant covectors and their signed composite metric. Within the explicitly adopted Einstein–Hilbert correspondence sector, independent static spherical variation selects $A^2 = B^2 = C_g/r$; matching the Newtonian limit fixes the integration constant $C_g = 2G_N M/c^2$. With one fixed intrinsic reference mass, this identifies the QGD inverse-amplitude normalization $|\alpha_G|^2 = \hbar c/(2G_N M m_{\rm ref})$. We retain the original four-mode Dirac spinor and give its consistently normalized, regular two-spin radial specialization. The exact static Killing-energy potential has a convergent classical Taylor series with no residual $\hbar$ dependence. Separately, a massless four-component Dirac loop in a specified Einstein-like metric effective theory supplies a conditional $\hbar$-dependent far-field $1/r^3$ term; massive Dirac matter has a screened far-field response. We demonstrate finite-source moments and an exact full-rank eight-sector signed natural-invariant representation. The normalization and classical correspondence are mathematically established in their stated effective sector; quantum-loop results are identified by their additional assumptions. Historical current, phase, and curved-Dirac tests remain preserved.

Status key. D: mathematical deduction from displayed hypotheses; A: explicitly adopted physical assumption/model; C: convention or matching definition; O: open or unverified QGD claim. A theorem labelled D does not assert that its assumptions have been microscopically derived.
How to read the results. The Newtonian limit fixes the exterior integration constant C_g, and α_G denotes one explicitly chosen inverse-strength reference normalization throughout; it is not an additional universal coupling. The exact square-root function below is the energy per unit rest mass of a *supported static probe*, not a universal scattering potential. The massless-Dirac b/r^3 term is calculated for an added gapless field within a specified Einstein-like effective-metric response; it is not silently inferred from the original massive Dirac spinor or from the classical normalization. The historical operator $\lambda\mathcal{Q}$ remains a separately identified trial, rather than an established quantum correction to the master metric.

1 Scope and exact dependencies

The gravitational constant G_N is an empirical parameter. Cavendish's torsion-balance experiment measured the gravitational attraction of laboratory masses (historically reported as the mean density of the Earth), providing a route to its determination. Einstein's geometric equations $G_{\mu\nu} = \kappa T_{\mu\nu}$ fix the universal coupling $\kappa = 8\pi G_N/c^4$ by recovering Newtonian gravitation. QGD uses the same correspondence methodology to fix an otherwise undetermined *source-dependent field-amplitude normalization* α_G; it does not identify α_G with Einstein's universal κ. We distinguish the assumed effective equations, their mathematical consequences, the one empirical matching condition, and conditional quantum predictions.

Take $G_N, M, c, \hbar, m_f, m_t, m_{\mathrm{ref}} > 0$, signature $(-+++)$, and write $x^0 = ct$. For stationary exterior statements let $r > r_s$ with $r_s = 2G_N M/c^2$. The Dirac angular quantum number will be called κ_D so that it is never confused with a gravitational coupling coefficient.

Chapter 1 input (local/coherent mode construction). The original four-component ansatz, with the local positive/negative-energy basis conventions made explicit, is

$$\boxed{\Psi(x) = \frac{\alpha_G}{r}\left[(\psi_0 u_\uparrow + \psi_1 u_\downarrow)e^{-\mathrm{i}S_+/\hbar} + (\psi_2 v_\uparrow + \psi_3 v_\downarrow)e^{+\mathrm{i}S_-/\hbar}\right].} \tag{1}$$

For a common local real momentum $\boldsymbol{p}$, let $\Delta = E_{\mathrm{loc}} + m_f c^2$ and $\boldsymbol{\zeta} = c\boldsymbol{p}/\Delta$. The four columns are

$$\begin{aligned} u_\uparrow &= \begin{pmatrix} 1 \\ 0 \\ \zeta_z \\ \zeta_x + \mathrm{i}\zeta_y \end{pmatrix}, \qquad u_\downarrow = \begin{pmatrix} 0 \\ 1 \\ \zeta_x - \mathrm{i}\zeta_y \\ -\zeta_z \end{pmatrix}, \\ v_\uparrow &= \begin{pmatrix} -\zeta_z \\ -\zeta_x - \mathrm{i}\zeta_y \\ 1 \\ 0 \end{pmatrix}, \quad v_\downarrow = \begin{pmatrix} -\zeta_x + \mathrm{i}\zeta_y \\ \zeta_z \\ 0 \\ 1 \end{pmatrix}. \end{aligned} \tag{2}$$

The physical Dirac lower/upper block ratio is

$$\boxed{\mathsf{R} = \frac{c\boldsymbol{\tau}\cdot\boldsymbol{p}}{E_{\mathrm{loc}} + m_f c^2} = \boldsymbol{\tau}\cdot\boldsymbol{\zeta}.} \tag{3}$$

The ansatz and ratio are exact local algebra for the stated on-shell spinors; turning arbitrary variable envelopes into a global Dirac solution requires the Dirac differential equations, WKB/spin-connection corrections, and boundary data. A general many-body density matrix is not automatically one coherent four-spinor.

Chapter 2 input (complete gravitational proposal). We preserve the full master metric

$$\boxed{g_{\mu\nu} = T^a{}_\mu T^b{}_\nu \left[\eta_{ab} - \sum_{s=1}^{N} \varepsilon_s \sigma_a^{(s)} \sigma_b^{(s)} - \lambda \mathcal{Q}_{ab}\right], \qquad \varepsilon_s = \pm 1.} \tag{4}$$

The promoted covectors $\sigma_\mu^{(s)}$ are the QGD *gravitational natural invariants*; they must not be identified componentwise with the local momentum ratios ζ_i. The promotion is a physical QGD hypothesis (O). The derivative term $\lambda\mathcal{Q}$ has not been microscopically matched and is excluded *only* from the named classical benchmark below. We do not assume it is identically zero in the unrestricted theory.

Effective classical benchmark (A). We explicitly adopt

$$S_0[g, \Psi] = \frac{c^3}{16\pi G_N}\int \mathrm{d}^4x\,\sqrt{-g}\,R + S_D[g, \Psi], \tag{5}$$

with $x^0 = ct$ and the overall measure/conversion convention understood so the vacuum variational equations are $G_{\mu\nu} = 0$ (equivalently set $c = 1$ during action variation). The Einstein–Hilbert action and minimal covariant Dirac coupling are *inputs*, not deduced from the flat microscopic Dirac action. Restrict to fixed reference tetrad, $\lambda = 0$, one positive signed mass sector, and stationary spherical symmetry. No conclusion below is asserted for all signed sectors or arbitrary tetrad variations.

2 The spherical natural-invariant field equations

Choose sufficiently smooth real functions A, B on a connected exterior radial interval and define

$$\sigma = A(r)\mathrm{d}x^0 + B(r)\mathrm{d}r, \qquad F = 1 - A^2, \qquad D = 1 + B^2 - A^2. \tag{6}$$

Assume $F > 0, D > 0$ and nonzero AB on the branch under examination. The classical specialization of Eq. (4) gives

$$\mathrm{d}s^2 = -F(\mathrm{d}x^0)^2 + 2AB\mathrm{d}x^0\mathrm{d}r + (1 + B^2)\mathrm{d}r^2 + r^2\mathrm{d}\Omega^2. \tag{7}$$

A local, static coordinate redefinition removes the cross term:

$$\boxed{\mathrm{d}s^2 = -F\mathrm{d}\tau^2 + \frac{D}{F}\mathrm{d}r^2 + r^2\mathrm{d}\Omega^2.} \tag{8}$$

The time coordinate is normalized at infinity. In the variational reduction, any field-dependent coordinate redefinition and its boundary variation must be handled consistently. Equivalently, one may perform the geometric computation in the original chart. The symmetry-restricted vacuum equations are sufficient for this particular nondegenerate two-function sector, not for the unrestricted master metric.

Lemma 2.1 (Exact scalar-curvature and boundary identity (D))**.** *For Eq. (8), with primes denoting* $\mathrm{d}/\mathrm{d}r$*, direct Levi–Civita computation gives*

$$R = -\frac{F''}{D} + \frac{F'D'}{2D^2} - \frac{4F'}{rD} + \frac{2FD'}{rD^2} + \frac{2}{r^2} - \frac{2F}{r^2D}. \tag{9}$$

The exact identity

$$\boxed{r^2\sqrt{D}\,R = 2\left[\frac{(rF)'}{\sqrt{D}} + \sqrt{D}\right] + \frac{\mathrm{d}}{\mathrm{d}r}\left[-\frac{r^2F'}{\sqrt{D}} - \frac{4rF}{\sqrt{D}}\right]} \tag{10}$$

holds without imposing field equations.

Proof. Compute the Christoffel symbols of Eq. (8), contract the Riemann tensor to obtain Eq. (9), and expand the total derivative in Eq. (10). All terms cancel identically after the product rule. In deriving equations of motion, one must either include the appropriate gravitational boundary term or specify fixed boundary variations so the total derivative does not change the bulk Euler–Lagrange equations. □

Proposition 2.2 (Independent natural-invariant variation (D|A))**.** *Under the classical benchmark, consistent boundary variation and* $AB \neq 0$*, the reduced bulk Lagrangian*

$$L_{\mathrm{red}} \propto \frac{(rF)'}{\sqrt{D}} + \sqrt{D} \tag{11}$$

has the independent radial Euler–Lagrange equations (up to a common nonzero normalization)

$$\boxed{\mathcal{E}_A = -\frac{A(rB^2)'}{D^{3/2}} = 0, \qquad \mathcal{E}_B = \frac{B\left[B^2 + r(A^2)'\right]}{D^{3/2}} = 0.} \tag{12}$$

With asymptotic flatness, they select

$$\boxed{A(r)^2 = B(r)^2 = \frac{C_g}{r}, \qquad C_g > 0.} \tag{13}$$

Proof. Treat A and B as independent variables before imposing equality of their profiles. Euler–Lagrange differentiation of Eq. (11) gives Eq. (12). Because $A, B \neq 0$, $(rB^2)' = 0$ and $B^2 + r(A^2)' = 0$. On a connected interval $rB^2 = C_g$; integrating $(A^2)' = -C_g/r^2$ yields $A^2 = C_g/r + C_0$. Asymptotic flatness and $A^2 \to 0$ imply $C_0 = 0$. Direct substitution in the full vacuum Einstein tensor gives zero for $r > 0$. The branch $A = 0$ or $B = 0$, source terms, and arbitrary tetrad/sector variations are not covered. □

Remark 2.3 (Scope). The result Eq. (13) is obtained from the *adopted* Einstein–Hilbert dynamics in a selected natural-invariant parametrization. It demonstrates a nontrivial exact gravitational *representation*; it is not a microscopic derivation of those dynamics from Chapter 1 and not a theorem for nonzero $\lambda\mathcal{Q}$.

3 Newtonian correspondence and the normalization of α_G

The positive nondegenerate branch of Proposition 2.2 has $g_{00} = -(1 - C_g/r)$. In the slow-motion, weak-field regime, $g_{00} = -(1 + 2\Phi_N/c^2) + O(c^{-4})$, and therefore

$$\Phi_N(r) = -\frac{c^2 C_g}{2r}, \qquad a_r = -\frac{\mathrm{d}\Phi_N}{\mathrm{d}r} = -\frac{c^2 C_g}{2r^2}. \tag{14}$$

The empirically established Newtonian limit for the source mass M fixes

$$\boxed{C_g = \frac{2G_N M}{c^2} =: r_s.} \tag{15}$$

This matches a single integration constant. It does not require a first-principles calculation of the numerical value of G_N.

Definition 3.1 (QGD gravitational normalization)**.** *Fix one intrinsic reference mass $m_{\mathrm{ref}} > 0$, independent of the mass or identity of any external probe, and the reduced Compton length $\bar{\lambda}_{\mathrm{ref}} = \hbar/(m_{\mathrm{ref}}c)$. Define the dimensionless normalization α_G through the* inverse natural-invariant strength

$$\boxed{C_g := \frac{\bar{\lambda}_{\mathrm{ref}}}{|\alpha_G|^2}.} \tag{16}$$

The reference mass is fixed throughout any comparison of source systems. In the specified positive spherical branch, this is equivalently

$$\frac{1}{|\alpha_G|^2} = \frac{1}{\bar{\lambda}_{\mathrm{ref}}} \lim_{r\to\infty} rA(r)^2. \tag{17}$$

For general signed-sector representations, C_g is identified using the asymptotic Killing metric/physical mass, not one preferred covector.

Theorem 3.2 (One-step Newtonian calibration of α_G)**.** *Given the field equations and boundary assumptions of Proposition 2.2, the definition (16) and Newtonian correspondence (15) fix the positive magnitude uniquely:*

$$\boxed{|\alpha_G|^2 = \frac{\hbar c}{2G_N M m_{\mathrm{ref}}}, \qquad \alpha_G = \sqrt{\frac{\hbar c}{2G_N M m_{\mathrm{ref}}}} \quad \textit{(positive-real convention).}} \tag{18}$$

Consequently the classical natural-invariant strength satisfies

$$\boxed{A(r)^2 = B(r)^2 = \frac{\hbar}{m_{\mathrm{ref}} c |\alpha_G|^2 r} = \frac{2G_N M}{c^2 r}.} \tag{19}$$

Proof. Equate the two expressions for C_g in (15) and (16) and solve for $|\alpha_G|^2$. Since all masses and couplings are positive, its magnitude is fixed. The phase is not determined by a real gravitational matching condition; choose the displayed positive representative. Substitute into $A^2 = B^2 = C_g/r$ to obtain (19). □

Remark 3.3 (Physical interpretation). $\kappa = 8\pi G_N/c^4$ is a universal coefficient of Einstein's field equations; α_G here parametrizes the inverse amplitude of a given source's natural-invariant exterior. They have analogous correspondence calibrations, not the same physical role. The historical phase $e^{i\pi/4}$ is not fixed by Newtonian matching. We use the *same symbol* α_G as the common multiplier of the Chapter 1 four-mode ansatz, adopting the corresponding reference-mode normalization convention; it is not recalibrated for each state or each power of $1/r$.

4 The complete normalized QGD spinor and regular ψ_0

Proposition 4.1 (Calibrated four-mode representation)**.** *With the phase convention fixed in Theorem 3.2, the original four-mode spinor* (1) *takes the explicit form*

$$\boxed{\Psi(x) = \frac{1}{r}\sqrt{\frac{\hbar c}{2G_N M m_{\mathrm{ref}}}}\big[(\psi_0 u_\uparrow + \psi_1 u_\downarrow)e^{-iS_+/\hbar} + (\psi_2 v_\uparrow + \psi_3 v_\downarrow)e^{+iS_-/\hbar}\big].} \tag{20}$$

The normalized mode coefficients ψ_a specify the chosen physical Dirac state and satisfy its differential equation and conserved-charge normalization; the equality of the prefactors is a fixed reference convention. It is not a second calibration of gravity.

For the special flat spherical coherent-mode reference with common real local momentum and angularly independent norm of the bracket $\mathcal{X}$,

$$\boxed{N_{\mathrm{wave}} = 4\pi|\alpha_G|^2\int_0^\infty \mathcal{X}^\dagger\mathcal{X}\,dr, \qquad \mathcal{X}^\dagger\mathcal{X} = (1+|\boldsymbol{\zeta}|^2)\sum_{a=0}^{3}|\psi_a|^2.} \tag{21}$$

This reference formula uses a coherent-spinor norm; a general many-fermion state is specified by a density operator and occupations. On a curved spatial hypersurface the conserved Dirac charge is the corresponding covariant $\int_\Sigma n_\mu J^\mu d\Sigma$, not a naive Euclidean integral.

Proposition 4.2 (Regular interacting spherical specialization)**.** *For a stationary $j = \frac{1}{2}$ spin-orbital in a specified smooth static source model, let U, V be the physical upper and lower radial functions, normalized by $4\pi\int_0^\infty r^2(U^2+V^2)dr = 1$ per occupied orbital. For a normalized Pauli spinor χ_A and the Pauli matrices $\boldsymbol{\tau}$,*

$$\boxed{\Psi_A(t,\mathbf{x}) = \frac{\alpha_G}{r}e^{-i\varepsilon t/\hbar}\begin{pmatrix}\psi_0(r)\chi_A\\ i\psi_L(r)(\boldsymbol{\tau}\cdot\hat{\mathbf{r}})\chi_A\end{pmatrix}, \quad \psi_0 = \frac{rU}{\alpha_G}, \quad \psi_L = \frac{rV}{\alpha_G}.} \tag{22}$$

Then $4\pi|\alpha_G|^2\int_0^\infty(|\psi_0|^2+|\psi_L|^2)dr = 1$. For bounded smooth central potentials on the regular $s_{1/2}$ branch, $U(r) = U(0) + O(r^2)$ and $V(r) = O(r)$, hence

$$\boxed{\psi_0(0) = 0, \quad \psi_L(0) = 0, \quad \psi_0'(0) = \frac{U(0)}{\alpha_G}, \qquad \psi_L(r) = O(r^2).} \tag{23}$$

Thus the physical upper spinor may be finite and nonzero at the origin, even though the reduced *ψ_0 vanishes there. The radial shape and $U(0)$ are fixed by the selected source Dirac problem, not by the Newtonian calibration alone.*

Proof. Substitute the definitions of ψ_0, ψ_L into the displayed spinor and normalization integral. Regularity follows by the standard smooth $s_{1/2}$ radial Frobenius expansion. □

Remark 4.3 (Interacting source benchmark). QGD–080/086 explicitly supply two spin-paired, self-consistent direct-Hartree Dirac orbitals in a *prescribed scalar confining well*, rather than one pure spinor representing an unpolarized two-particle state. The orbitals have $\varepsilon \simeq 0.471092477$ in their stated natural units and per-orbital numerical charge 1.000000499. Their complete gravitational source would include the energy/stress of the confinement apparatus; that contribution has not been computed in this benchmark.

5 Exact static Killing-energy potential

Definition 5.1 (Operational meaning of Φ_{stat})**.** *For a pointlike test body of rest mass $m_t > 0$* held static *in the exterior of Eq. (8), define*

$$E_\infty(r) = m_t c^2 \sqrt{-g_{00}(r)}, \qquad \Phi_{\text{stat}}(r) := \frac{E_\infty(r) - m_t c^2}{m_t}. \tag{24}$$

The Killing time is normalized at infinity and $\Phi_{\text{stat}}(\infty) = 0$.

Theorem 5.2 (Exact normalized static potential (D|A,C))**.** *For $r > r_s$, the definition Eq. (24) and Eq. (15) imply*

$$\boxed{\Phi_{\text{stat}}(r) = c^2 \left[\sqrt{1 - \frac{2G_N M}{c^2 r}} - 1 \right].} \tag{25}$$

Writing $x = r_s/r \in (0,1)$, the equivalent rationalized expression is

$$\Phi_{\text{stat}} = -\frac{c^2 x}{1 + \sqrt{1-x}}. \tag{26}$$

Proof. Insert $g_{00} = -(1 - r_s/r)$ into Eq. (24); multiplying $\sqrt{1-x} - 1$ by $(\sqrt{1-x}+1)/(\sqrt{1-x}+1)$ proves Eq. (26). No quantum correction and no Taylor approximation enter this exact identity. □

Remark 5.3 (Three inequivalent "potentials"). The weak-field metric coefficient $\Phi_N = -G_N M/r$ describes Newton's limiting force and is *exactly linear* in C_g/r for this Schwarzschild chart. The quantity Φ_{stat} is the position-dependent static Killing-energy difference per unit mass. The radial effective potential obtained by eliminating a component of a Dirac partial wave is a third, representation-dependent object. They coincide only in the stated leading Newtonian regime and must not be interchanged beyond it.

5.1 Optional notation $\psi_{\text{grav}}(r)$: not a Dirac wavefunction

Definition 5.4 (Gravitational potential expressed using ψ)**.** *To distinguish the gravitational scalar from the Chapter 1 quantum amplitudes $\psi_0, \ldots, \psi_3$ and the four-component spinor Ψ, define*

$$\boxed{\psi_{\text{grav}}(r) := \Phi_{\text{stat}}(r) = c^2 \left(\sqrt{1 - \frac{2G_N M}{c^2 r}} - 1 \right).} \tag{27}$$

The notation ψ_{grav} is an alias *for the operationally defined static Killing-energy potential per unit test mass, not an identification with the radial quantum wavefunction and not a new gravitational degree of freedom. It has units of squared velocity, whereas a normalized Dirac wavefunction has probability-amplitude dimensions. Its exact classical expansion is proved in Theorem 6.1; a possible quantum correction must be derived from newly specified source or field dynamics and is not fixed by notation.*

6 All-order expansion, force and rigorous error bound

Theorem 6.1 (Convergent expansion (D)). *Let $0 < x = r_s/r < 1$ and define*

$$a_n = \frac{\binom{2n}{n}}{4^n(2n-1)}, \qquad n \geq 1. \tag{28}$$

Then $a_n > 0$ and $a_{n+1}/a_n = (2n-1)/(2n+2) < 1$. The binomial series gives

$$\boxed{\Phi_{\text{stat}}(r) = -c^2 \sum_{n=1}^{\infty} a_n \left(\frac{r_s}{r}\right)^n.} \tag{29}$$

In particular,

$$\boxed{\begin{aligned}\Phi_{\text{stat}}(r) = &-\frac{G_N M}{r} - \frac{G_N^2 M^2}{2c^2 r^2} - \frac{G_N^3 M^3}{2c^4 r^3} \\ &- \frac{5G_N^4 M^4}{8c^6 r^4} + \mathcal{R}_4(r),\end{aligned}} \tag{30}$$

Equivalently, in the optional notation just defined,

$$\boxed{\psi_{\text{grav}}(r) = -\frac{G_N M}{r} - \frac{G_N^2 M^2}{2c^2 r^2} - \frac{G_N^3 M^3}{2c^4 r^3} - \frac{5G_N^4 M^4}{8c^6 r^4} + \mathcal{R}_4(r).} \tag{31}$$

with the explicit bound

$$\boxed{|\mathcal{R}_4(r)| \leq \frac{7c^2}{256}\frac{x^5}{1-x}.} \tag{32}$$

Proof. Apply the generalized binomial theorem to $\sqrt{1-x}$, valid for $|x| < 1$. The first five positive a_n are $1/2, 1/8, 1/16, 5/128, 7/256$. The ratio formula shows $a_n \leq a_5$ for $n \geq 5$, so $|\mathcal{R}_4| \leq c^2 a_5 \sum_{n\geq 5} x^n$, proving Eq. (32). Substitute $r_s = 2G_N M/c^2$ to obtain Eq. (30). The series is absolutely convergent for every $r > r_s$. □

Corollary 6.2 (Static support force (D)). *The negative radial derivative of the supported body's potential energy is*

$$\boxed{F_{\text{stat}}(r) := -m_t \frac{\mathrm{d}\Phi_{\text{stat}}}{\mathrm{d}r} = -\frac{G_N M m_t}{r^2}\left(1 - \frac{r_s}{r}\right)^{-1/2}.} \tag{33}$$

Its weak-field expansion begins

$$F_{\text{stat}} = -\frac{G_N M m_t}{r^2}\left[1 + \frac{G_N M}{c^2 r} + \frac{3G_N^2 M^2}{2c^4 r^2} + \mathcal{O}(r^{-3})\right]. \tag{34}$$

Proof. Differentiate Eq. (25) and use the ordinary binomial series for $(1-x)^{-1/2}$. □

Remark 6.3 (Force interpretation). F_{stat} is a derivative of a supported body's energy with respect to areal radius; it is not the magnitude of the proper acceleration force read by a local static observer. The latter has an additional radial-metric/redshift conversion. Thus even the exact classical r^{-2}-corrected series is not automatically a universal force acting on every free-falling quantum mode.

7 Classical cancellation of $\hbar$

Using the calibrated α_G in the inverse-strength convention yields

$$\boxed{\frac{\hbar}{m_{\mathrm{ref}} c|\alpha_G|^2} = \frac{2G_N M}{c^2} = C_g.} \tag{35}$$

Every coefficient of the all-order classical expansion (29) is thus independent of the reference mass and of $\hbar$; its physics is the classical Schwarzschild static Killing energy. This cancellation is *not* a statement that quantum matter has no gravitational effects. Quantum corrections must be calculated separately, as below.

8 First quantum matter-loop correction in the effective metric sector

Here we retain the Newton-calibrated G_N and α_G of Theorem 3.2 without further fitting. In addition to the adopted Einstein-like dynamical metric, assume one *massless four-component Dirac field* coupled covariantly to it. This is a specified weak-field one-loop benchmark, distinct from the massive scalar-contact Dirac source whose nonlocal vacuum response is exponentially suppressed.

Proposition 8.1 (Conditional one-loop source response)**.** *In the flat-background weak-field effective-metric conventions of QGD–083/084, the properly normalized massless-Dirac spin-two spectral response has $\rho_2(s) = s^2/(80\pi)$, and the Fourier transform of the first-order static sourced temporal metric perturbation takes the form*

$$\boxed{\widetilde{H}(\mathbf{k}) = \frac{4\pi C_g \widetilde{f}(\mathbf{k})}{k^2}\left[1 - \frac{\ell_P^2 k^2}{15\pi}\log\left(\frac{k^2}{\mu^2}\right)\right] + O(\ell_P^4), \qquad \ell_P^2 = \frac{G_N\hbar}{c^3}.} \tag{36}$$

Here f is a normalized stationary source shape and μ is a renormalization scale. The coefficient is that of the adopted massless-Dirac plus Einstein-like metric benchmark; the full nonlinear QGD signed-sector quantum dynamics are not assumed solved by it. For a point source, at separated exterior points,

$$\boxed{\delta H(r) = \frac{b}{r^3}, \qquad b = \frac{2C_g\ell_P^2}{15\pi} = \frac{4G_N^2 M\hbar}{15\pi c^5}.} \tag{37}$$

Proof. The conventionally subtracted massless spin-two loop supplies a $k^4\log(k^2/\mu^2)$ metric Hessian, with the Dirac vertex normalization and seagull/contact prescription summarized by QGD–084. Expand the calibrated metric propagator through first order in the loop: it gives (36). Using the distributional identity

$$\mathcal{F}^{-1}_{\mathbf{k}\to\mathbf{x}}\left[\log(k^2/\mu^2)\right] = -\frac{1}{2\pi r^3}, \qquad r > 0, \tag{38}$$

with contact terms confined to the source, gives (37). This is a result for the *specified* effective-metric observable and normalization convention; it is not obtained by Taylor-expanding a complex wave phase or modifying α_G. □

Remark 8.2 (Massive matter versus massless matter). For a massive Dirac loop, the two-fermion spectral cut begins at $2m$; the long-distance response is exponentially screened and does not give a nonzero asymptotic constant b of the form (37). The power-law coefficient above requires the stated gapless matter sector.

Proposition 8.3 (Finite-source one-loop exterior)**.** *If $f(r) \geq 0$ is normalized, spherically symmetric, and supported in $0 \leq r \leq R$, the weak-field first-order loop correction outside that source is*

$$\boxed{\delta H(r) = \frac{b}{r^3}\int_0^R \frac{4\pi s^2 f(s)}{1 - s^2/r^2}\, ds = \frac{b}{r^3}\left[1 + \frac{\mu_2}{r^2} + \frac{\mu_4}{r^4} + \cdots\right], \quad r > R,} \tag{39}$$

where $\mu_j = 4\pi \int_0^R s^{j+2} f(s) ds$. Thus the leading coefficient b depends on total mass but not on the internal source radius, while source-structure effects begin at r^{-5}.

Proof. Convolve the separated-point kernel $b/|\mathbf{x} - \mathbf{y}|^3$ with the radial source, carry out the angular integral and expand the positive geometric series in s^2/r^2 for $r > R$. □

Theorem 8.4 (Conditional quantum-corrected Taylor hierarchy)**.** *If the exterior temporal redshift is described by the same static potential generator as Theorem 5.2, now with $H = C_g/r + b/r^3 + b\mu_2/r^5 + \cdots$, then formal expansion yields*

$$\boxed{\begin{aligned}\Phi_{\mathrm{stat}}(r) = -c^2\Bigg[&\frac{C_g}{2r} + \frac{C_g^2}{8r^2} + \frac{1}{r^3}\left(\frac{C_g^3}{16} + \frac{b}{2}\right) \\ &+ \frac{1}{r^4}\left(\frac{5C_g^4}{128} + \frac{C_g b}{4}\right) \\ &+ \frac{1}{r^5}\left(\frac{7C_g^5}{256} + \frac{3C_g^2 b}{16} + \frac{b\mu_2}{2}\right) + O(r^{-6})\Bigg].\end{aligned}} \tag{40}$$

Its leading quantum contribution is

$$\boxed{\delta\Phi^{(3)}_{\mathrm{stat}}(r) = -\frac{c^2 b}{2r^3} = -\frac{2G_N^2 M\hbar}{15\pi c^3 r^3}.} \tag{41}$$

Proof. Insert the stated H into $c^2(\sqrt{1-H} - 1)$ and collect powers of $1/r$. The terms involving b are algebraic deductions from the redshift-generator assumption. □

Remark 8.5 (Observable and perturbative order). Equation (36) computes a first-order, weak-field metric response. Only the leading quantum far-field coefficient is provided by that linearized one-loop calculation. The mixed $C_g b/r^4$ and higher corrections in (40) are *conditional nonlinear consequences of assuming* the same static Killing redshift generator for the corrected metric. They are not an independently solved quantum-corrected Einstein–Dirac spacetime. They must not be identified with coefficients of a scattering potential or a gauge-dependent radial wave-equation potential without an observable matching calculation.

9 Finite Dirac sources and the full signed natural-invariant sector

For two occupied spin-paired stationary orbitals ϕ_A of a specified Hartree–Dirac source, define the basis-independent occupied projector

$$\boxed{\mathcal{R}(\mathbf{x}, \mathbf{y}) = \sum_{A=1}^{2} \phi_A(\mathbf{x})\phi_A^\dagger(\mathbf{y}).} \tag{42}$$

Constant unitary changes of occupied-orbital basis leave $\mathcal{R}$ unchanged. The resulting number density, current, scalar condensate and kinetic/stress observables are therefore representation-independent. For the radial spin-paired source of Proposition 4.2, $n = 2(U^2 + V^2)$, $S = 2(U^2 - V^2)$ and $\mathbf{j} = 0$ even though the kinetic spatial tensor can be nonzero.

The *adopted* Newtonian/Einsteinian weak-field equation couples the *complete* conserved source energy density $\mathcal{E}_{\rm tot}$ to the static gravitational field:

$$\boxed{H_0(\mathbf{x}) = \frac{2G_N}{c^4}\int \frac{\mathcal{E}_{\rm tot}(\mathbf{y})}{|\mathbf{x}-\mathbf{y}|}\,d^3y, \qquad -\nabla^2 H_0 = \frac{8\pi G_N}{c^4}\mathcal{E}_{\rm tot}.} \tag{43}$$

For $\int \mathcal{E}_{\rm tot} d^3x = Mc^2$ this has $H_0 \sim C_g/r$; it does not introduce another value of α_G. The QGD–080 benchmark uses an externally prescribed binding well, so its Dirac-only energy profile serves as a source-*shape* test and is not the complete conserved stress-energy including confinement.

Theorem 9.1 (Exact full-rank signed metric section)**.** *In a fixed smooth reference coframe, let $h = g - \bar{g}$ be any real symmetric 4×4 metric correction. Use the eight signed one-form sectors of the metric-only specialization of the Chapter 2 master metric. Define the four positive and four negative rows by*

$$\boxed{P = sI_4 + \frac{h}{4s}, \qquad N = sI_4 - \frac{h}{4s}, \qquad s > 0.} \tag{44}$$

Then $P^TP - N^TN = h$ identically. Its derivative admits the right inverse $\delta P = \delta h/(4s)$, $\delta N = -\delta h/(4s)$ and therefore has rank ten on this section, including at $h = 0$.

Proof. For $h = h^T$, expand $P^TP - N^TN$; both quadratic h^2 terms cancel and the linear terms sum to h. Differentiating the same identity proves the displayed right inverse. □

Writing $S = \binom{P}{N}$ and $J = \mathrm{diag}(I_4, -I_4)$, we have $h = S^TJS$, invariant under pointwise transformations $S \mapsto O(x)S$ with $O^TJO = J$. Whenever the *full action and matter couplings depend on the signed fields only through the metric*, these are local $O(4,4)$ parametrization redundancies, not extra metric observables. At a full-rank section the source-and-field variations satisfy

$$\boxed{\frac{\delta\Gamma}{\delta\sigma} = 0 \quad\Longleftrightarrow\quad \frac{\delta\Gamma}{\delta g} = 0 \quad \text{(metric-only full-rank sector).}} \tag{45}$$

Derivative terms separately coupling to individual signed σ_μ sectors, including the unverified $\lambda\mathcal{Q}$, need not preserve this redundancy; their status belongs to the historical Chapter 4 field operator investigation, not the present verified effective metric-only sector.

10 The Chapter 1 Dirac-current bridge

For a single approximately outgoing radial positive-energy WKB branch, write

$$\Psi \simeq \frac{\alpha_G\psi_0(r)}{r}\begin{pmatrix}\chi\\ \dfrac{cp(r)}{\Delta(r)}(\boldsymbol{\tau}\cdot\hat{\boldsymbol{r}})\chi\end{pmatrix} e^{-\mathrm{i}[Et-W(r)]/\hbar}, \quad \chi^\dagger\chi = 1, \quad W' = p > 0. \tag{46}$$

The small lower components must be retained even in a nonrelativistic flux calculation. Direct Dirac-matrix multiplication gives

$$j^r = c\Psi^\dagger(\boldsymbol{\alpha}\cdot\hat{\boldsymbol{r}})\Psi = \frac{2c^2|\alpha_G\psi_0|^2}{r^2}\frac{p}{\Delta}. \tag{47}$$

For stationary spherical current conservation $\partial_r(r^2 j^r) = 0$,

$$\boxed{\dot{N} = 8\pi c^2\frac{|\alpha_G\psi_0(r)|^2 p(r)}{\Delta(r)}, \qquad |\alpha_G\psi_0(r)|^2 = \frac{\dot{N}\Delta(r)}{8\pi c^2 p(r)}.} \tag{48}$$

Equivalently, using $\lambda_{\mathrm{dB}} = h/p$ and $\mathcal{C}_J^2 = \dot{N}/(8\pi c^2 h)$,

$$\boxed{|\alpha_G \psi_0(r)|^2 = \mathcal{C}_J^2 \Delta(r) \lambda_{\mathrm{dB}}(r).} \tag{49}$$

This is an *amplitude/current* bridge, not a complex equality. No factor $e^{\pm 2\mathrm{i}S/\hbar}$ survives a modulus squared. It determines how an assigned current transports along an already specified dispersion relation; it does not derive a force profile from flat source-free Dirac kinematics. Here $\mathcal{C}_J$ is a flux parameter, not the gravitational integration constant C_g.

Remark 10.1 (Zero-flux source). A regular stationary, self-gravitating Dirac configuration may have zero net radial current and nonzero energy density. Dividing by $\dot{N}$ or applying Eq. (49) to such a source is invalid. The normalized two-spinor Einstein–Dirac source benchmark verifies self-consistent matter/metric dynamics under the adopted gravitational equations, but it does not make the historical geometric and spinor α_G numerically identical.

11 A controlled quantum radial wave example

Consider an external Newtonian gravitational potential $\Phi_N = -G_N M/r$, a nonrelativistic test mass $m_t > 0$, zero orbital angular momentum and zero *nonrelativistic* energy $\varepsilon = 0$. These are additional test-wave assumptions, not a microscopic derivation of gravity. The reduced radial Schrödinger function $u(r)$ in $\Psi \propto u(r)/r$ satisfies

$$-\frac{\hbar^2}{2m_t} u'' + m_t \Phi_N u = 0, \qquad u'' + \frac{2G_N M m_t^2}{\hbar^2 r} u = 0. \tag{50}$$

With $\beta = 2G_N M m_t^2/\hbar^2$ and $z = 2\sqrt{\beta r}$, an exact complex outgoing-basis solution is

$$\boxed{u_{\mathrm{out}}(r) = \mathcal{A}\sqrt{r}\, H_1^{(1)}(2\sqrt{\beta r}).} \tag{51}$$

Writing $u_{\mathrm{out}} = R \exp(\mathrm{i}W/\hbar)$ with $R \geq 0$, we obtain

$$R = |\mathcal{A}|\sqrt{r}\sqrt{J_1(z)^2 + Y_1(z)^2}, \quad W' = \frac{\hbar}{\pi r[J_1(z)^2 + Y_1(z)^2]}, \quad \boxed{R^2 W' = \frac{\hbar|\mathcal{A}|^2}{\pi}.} \tag{52}$$

The last identity is the exact nonrelativistic amplitude–momentum current law for this wave. At $z \gg 1$, $R \propto r^{1/4}$, agreeing with the leading flux/WKB envelope. The wave is not square-integrable at infinity as a stationary outgoing scattering state, and its Y_1 component produces an origin singularity in the full u/r field. A regular bound source needs different boundary conditions.

Separating the real and imaginary parts of Eq. (50) yields the state-dependent quantum-potential contribution

$$\frac{W'^2}{2m_t} + m_t \Phi_N + Q = 0, \qquad Q := -\frac{\hbar^2}{2m_t}\frac{R''}{R}. \tag{53}$$

In the controlled asymptotic regime $z \gg 1$ the first term in Q is $3\hbar^2/(32 m_t r^2)$, giving a corresponding radial hydrodynamic term $-Q' \simeq +3\hbar^2/(16 m_t r^3)$. This is quantum *wave dynamics in a prescribed classical potential*, not a new gravitational source law or a universal correction to Φ_{stat}.

12 Curved Dirac dynamics and the historical quantum-force test

12.1 Curved-space propagation: an additional, explicit model input

Chapter 2's smooth Lorentzian composite metric admits a local orthonormal frame and corresponding Clifford matrices. To propagate a test spinor, *adopt* the minimally coupled covariant Dirac equation with its spin connection,

$$(\mathrm{i}\hbar c\, \gamma^\mu \nabla_\mu - m_t c^2)\Psi = 0, \qquad \nabla_\mu = \partial_\mu + \Omega_\mu. \tag{54}$$

Using the classical Schwarzschild exterior and a fixed spherical tetrad/spinor-harmonic convention, let r_* denote tortoise radius $\mathrm{d}r_*/\mathrm{d}r = 1/F$, and let $\kappa_D \in \mathbb{Z} \setminus \{0\}$. A standard separated radial system is

$$\begin{aligned} \frac{\mathrm{d}f}{\mathrm{d}r_*} &= \frac{E + m_t c^2\sqrt{F}}{\hbar c}\, g - \frac{\kappa_D\sqrt{F}}{r} f, \\ \frac{\mathrm{d}g}{\mathrm{d}r_*} &= \frac{\kappa_D\sqrt{F}}{r} g + \frac{m_t c^2\sqrt{F} - E}{\hbar c} f. \end{aligned} \tag{55}$$

A local tetrad/spinor basis choice and radial rescalings are part of these hypotheses. Equation Eq. (55) is a separated *test-field* Dirac system, not the unrestricted self-gravitating four-mode QGD field equation.

12.2 An exact radial reduction and its weak-field coefficients

Let $\ell_C = \hbar/(m_t c)$, $x = r/\ell_C$, $a = r_s/\ell_C$, $e = E/(m_t c^2)$, $F = 1 - a/x$, $s = \sqrt{F}$, and $\mathrm{d}/\mathrm{d}x_* = F\mathrm{d}/\mathrm{d}x$. Define $B_e = e + s$ and $K = \kappa_D s/x$. The coupled system Eq. (55) becomes

$$f_*' = B_e g - K f, \qquad g_*' = K g + (s - e) f. \tag{56}$$

On a domain where $F > 0$ and $B_e > 0$, eliminating g gives

$$f_{**}'' - \frac{B_{e,*}'}{B_e} f_*' + \left[K_*' - \frac{B_{e,*}'}{B_e} K - K^2 + e^2 - F \right] f = 0. \tag{57}$$

Write $P = F_x/F - (B_e)_x/B_e$ and $q = [K_*' - \frac{B_{e,*}'}{B_e} K - K^2 + e^2 - F]/F^2$; with $u = \sqrt{F/B_e}\, f$, the Liouville transformation yields

$$\boxed{u'' + \mathcal{V}_{\kappa_D}(x, a, e) u = 0, \qquad \mathcal{V}_{\kappa_D} = q - \frac{1}{2} P_x - \frac{1}{4} P^2.} \tag{58}$$

At threshold energy $e = 1$, Taylor expansion at fixed $x > 0$ gives

$$\boxed{\mathcal{V}_{\kappa_D}(x, a, 1) = -\frac{\kappa_D(\kappa_D + 1)}{x^2} + \frac{a}{x} + \frac{3 - 4\kappa_D^2 - \kappa_D}{4} \frac{a}{x^3} + \mathcal{O}(a^2).} \tag{59}$$

This identity has been checked independently using exact symbolic differentiation, including the flat limit. The coefficient of a/x^3 is $\gamma_{\kappa_D} := (3 - 4\kappa_D^2 - \kappa_D)/4$, so $\gamma_{-1} = 0$ and $\gamma_{+1} = -1/2$. For the $\kappa_D = -1$ channel, the next expansion is

$$\boxed{\mathcal{V}_{-1}(x, a, 1) = \frac{a}{x} + \frac{2a^2}{x^2} + \frac{a^2}{64x^4} + \mathcal{O}(a^3).} \tag{60}$$

The a/x^3 term, when rewritten as a *one-dimensional effective potential coefficient*, produces a formal contribution proportional to $GM\hbar^2/(m_t c^2 r^3)$; its derivative has r^{-4} scaling. For $\kappa_D = -1$ that particular linear-G term vanishes. For $\kappa_D = +1$ the formal differentiated term points outward with magnitude $3GM\hbar^2/(2m_t c^2 r^4)$.

Remark 12.1 (Strict interpretation gate). The Liouville transformation, angular channel, tetrad and choice of radial variable affect the one-dimensional *radial potential coefficient*. The negative derivative of that coefficient is *not* automatically an invariant observable, a force measured by a detector, or a source-independent correction to $g_{\mu\nu}$. In particular Eq. (59)–(60) do not derive a quantum correction to the QGD gravitational natural invariant or to Eq. (25). A physical force claim requires an operational observable and a calculation of its dynamics, including state and source dependence.

12.3 Historical 9/2 claim: exact algebraic audit

The archived candidate is commonly displayed as an attractive radial force

$$F_{r,\mathrm{hist}} = -\frac{G_N M m_t}{r^2}\left[1 + \frac{9}{2}\left(\frac{\hbar}{m_t c r}\right)^2 + \cdots\right]. \tag{61}$$

It originally arose by expanding $e^{2\mathrm{i}m_t cr/\hbar}$ and inverting individual Taylor terms as though they were separately invertible contributions to one reciprocal. This step is algebraically invalid:

$$\frac{1}{\sum_n a_n} \neq \sum_n \frac{1}{a_n}, \qquad \frac{1}{e^{2\mathrm{i}m_t cr/\hbar}} = e^{-2\mathrm{i}m_t cr/\hbar}. \tag{62}$$

Moreover, the original cubic term gives $E_3 = +3G_N M\hbar^2/(2m_t c^2 r^3)$ under its formal term-by-term inversion, whose negative derivative points *outward*, opposite to the sign in Eq. (61). The small-phase regime $r \ll \hbar/(m_t c)$ also conflicts with treating the proposed correction as small for $r \gg \hbar/(m_t c)$. The historical number 9/2 is consequently *not a derived universal quantum-gravitational coefficient.* The curved Dirac test Eq. (59) does not rescue it.

12.4 What would count as a quantum-corrected gravitational potential?

To modify Φ_{stat} itself rather than quantum propagation in a fixed background, QGD needs independently specified gravitational effective dynamics and a source, for example a controlled renormalized effective action $\Gamma[g(\sigma), \Psi]$ with justified higher-derivative or nonlocal terms, or a quantum-matter stress tensor $\langle T_{\mu\nu}\rangle$ with specified state and renormalization. A term $\lambda\mathcal{Q}$ or a kinetic operator $\propto \hbar^2(\nabla\sigma)^2$ must not be inserted as though it follows from Chapters 1–2 until its covariance, dimension, microscopic provenance and full variation are established. In a sufficiently general low-energy parameterization one may write a *candidate*

$$\Phi_{\mathrm{phys}} = \Phi_{\mathrm{stat}} + \delta\Phi_{\mathrm{source}}(r;\hbar) + \delta\Phi_{\mathrm{grav}}(r;\hbar), \tag{63}$$

but only the explicitly defined massless-Dirac effective-metric benchmark of Section 8 has the calculated b of (37); no arbitrary universal c_Q is asserted for other fields. Equation Eq. (63) remains a useful separation of physical effects. For instance, dimensional analysis permits the clearly *unverified* candidate

$$\psi_{\mathrm{grav}}(r;\hbar) = \psi_{\mathrm{grav}}(r) - c_Q \frac{G_N M\hbar^2}{m_{\mathrm{ref}}^2 c^2 r^3} + \cdots, \tag{64}$$

where c_Q is an undetermined dimensionless coefficient for this separate historical trial form, not the calculated massless-Dirac coefficient b. No value or sign is implied by the Dirac-current bridge or the adopted Einstein–Hilbert vacuum equations. If m_{ref} is a test-particle mass, the proposed correction would not be a universal test-body-independent metric potential. A proposed force coefficient must be obtained by differentiating a *physically justified* energy, not by selecting Taylor-series terms.

13 Proof-status ledger and verification obligations

Statement	Status	Essential qualification
Dirac lower/upper ratio and four columns	D	Local on-shell common momentum; globally WKB only
Micro-to-σ_μ promotion	O	Foundational hypothesis, not from spinor ratios alone

Statement	Status	Essential qualification
Full signed master metric	A	Definition; derivative term unmatched
Classical Einstein–Hilbert action	A	Adopted effective dynamics
Spherical metric and curvature identities	D|A	Fixed tetrad; classical restricted sector
Two independent radial Euler–Lagrange equations	D|A	Boundary terms; $ABFD \neq 0$
$A^2 = B^2 = C_g/r$ exterior	D|A	Nondegenerate vacuum, asymptotic flatness
$C_g = 2G_N M/c^2$	C	Weak-field Newton matching
QGD $\|\alpha_G\|^2$	C	Newton calibration and fixed source-reference convention
Exact static Killing-energy potential	D|A,C	Supported probe and Killing time
Higher-order coefficients and error bound	D	$0 < r_s/r < 1$; classical terms
Dirac radial flux bridge	D|A	One outgoing WKB branch only
Exact zero-energy Newtonian Hankel radial wave	D|A	External Newtonian potential supplied
Curved Dirac radial expansion	D|A	Minimal coupling, tetrad, threshold E
Historical universal 9/2 quantum force	O	Invalid original reciprocal manipulation
Regular radial ψ_0 and full spinor	D|A,C	Specified normalized source state; no second gravitational fit
Massless Dirac b in effective metric	D|A	Additional massless species; linearized metric benchmark
Signed $O(4,4)$ section and rank	D|A	Full eight-sector metric-only model
Lean 4 kernel verification	O	Not executed; symbols/numerics are not Lean

Formal proof program. Lean 4+Mathlib formalization should target (L1) block metric identities and coordinate reparametrization; (L2) Ricci scalar in the stated smooth radial domain; (L3) the explicit total derivative and chosen boundary hypotheses; (L4) Euler–Lagrange derivatives with two independent fields; (L5) connected-interval radial integration and asymptotic conditions; (L6) Newtonian coefficient comparison; (L7) Killing-energy algebra; (L8) binomial-series coefficients, convergence and remainder; (L9) global spinor-rescaling invariance; (L10) current-bilinear identities and flux law under WKB hypotheses; (L11) Liouville transformation and Taylor coefficients with $F, B_e > 0$. Geometry/model assumptions must be recorded as hypotheses, not hidden axioms that assert the conclusions. Until those proofs compile, the precise phrase is *symbolically verified, Lean pending.*

14 Conclusion

The QGD classical effective sector permits a mathematically exact single-calibration correspondence: its independently varied nondegenerate spherical field equations give $A^2 = B^2 = C_g/r$, Newtonian matching gives $C_g = 2G_N M/c^2$, and the fixed reference-length definition yields $\alpha_G = \sqrt{\hbar c/(2G_N M m_{\mathrm{ref}})}$ in the positive convention. The same symbol is used consistently in the complete four-mode Dirac spinor with correspondingly normalized mode envelopes; a regular interacting two-spin specialization provides explicit ψ_0 and lower radial amplitudes. The classical static Killing-energy potential and its all-order Taylor coefficients follow without further fitting.

An independent massless-Dirac one-loop response in a specified Einstein-like effective metric adds a first-order quantum b/r^3 correction with $b = 4G_N^2 M\hbar/(15\pi c^5)$, with finite-source structure first entering at order r^{-5}. Higher mixed terms obtained from the square-root generator are stated

as conditional nonlinear extrapolations, rather than claimed exact one-loop-corrected spacetimes. The full signed eight-sector metric-only map admits an exact smooth full-rank section and equivalent variational equations modulo its internal local redundancy. The original current, historical phase and curved test-Dirac calculations are retained with their original domains of validity; the unverified independent natural-invariant derivative operators remain outside Chapter 3. Newtonian and Einsteinian correspondence are openly adopted where used: the present claims are results of that well-defined effective QGD formulation, not a prediction of the measured value of G_N.

A Dimensions and distinct constants

Quantity	SI dimensions
$G_N M/c^2$, r_s, C_g, $\bar{\lambda}_{\mathrm{ref}}$	length m
Φ_N, Φ_{stat}	$\mathrm{m}^2\mathrm{s}^{-2}$
α_G	dimensionless
$b = 2C_g\ell_P^2/(15\pi)$	length cubed m^3
$\dot{N}$	s^{-1}
$\mathcal{C}_J^2 = \dot{N}/(8\pi c^2 h)$	determined by flux bridge
Q	energy J
$\delta F_{\kappa_D}^{\mathrm{radial}}$	force N

The symbols C_g (geometrical integration constant) and $\mathcal{C}_J$ (Dirac wave flux scale) are unrelated. Neither may be replaced with the other. The calibrated α_G is used consistently as the QGD field-reference normalization in the original four-mode ansatz; the state amplitudes are normalized relative to that convention. The loop coefficient b has dimensions of length cubed.

B Reproducibility and source provenance

This surgical revision preserves the earlier accepted chapter material and adds QGD–082 (spinor calibration), QGD–083/084 (Dirac-loop spectral and vertex normalization), QGD–085 (finite-source and signed-sector equivalence), and QGD–086 (physical two-orbital Dirac source). Earlier QGD–062, 078/079, 083/084, 086–092 refer to the historical series of verification audits and are not to be confused with the new 2026 QGD–082–086 research sequence. Their symbolic verifiers check exactly stated algebraic identities; numerical benchmarks are tests of specified effective systems, not unrestricted existence or universality proofs. The source chapters have not been modified by this draft.